Rushda Jagtap

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PROFILE

B Tech CSE Sophomore with a focus in cyber adversarial analytics: the convergence of data science and cybersecurity that I treat not as two domains, but rather an art. I create adaptive solutions that understand what is normal to detect what rules cannot. Valuing utility over benchmarking, I explore machine learning security. I enjoy reading, sketching, and outdoors.

EDUCATION

MIT Academy of Engineering, Pune — **2025 – 2029 (expected)**
B. Tech Computer Science & Engineering (Data Science) | CGPA: 8.90/10 (Year 1) — *Pune, MH*

Amrita Vidyalayam, Navi Mumbai — ***2023 – 2025***
Class XII | Percentage: 86.40% — *Navi Mumbai, MH*

Amrita Vidyalayam, Navi Mumbai — ***2023 – 2025***
Class X | Percentage: 95.40% — *Navi Mumbai, MH*

PROJECTS

NEITH: AI-Powered Network Intrusion Detection System — GitHub
Python, Scapy, GraphSAGE (PyTorch Geometric), Flask, Next.js, SQLite

- Designed a GraphSAGE GNN modeling live traffic as a graph to flag unseen attack patterns; trained on CICIDS 2017 DDoS (83.5% accuracy); real-time pipeline: 28 CICIDS-aligned features per IP every 10s from live packet capture, with reverse DNS labeling and device role inference; mapped anomalies to 9 MITRE ATT&CK techniques via 90%-coverage conformal prediction intervals and ADWIN concept-drift detection; deployed full-stack live dashboard with D3.js network graph and persistent SQLite alert history.

Sanchay-AI: AI-Powered MPLADS Audit Intelligence Platform — GitHub
Python, FastAPI, React, TypeScript, scikit-learn, PostgreSQL, SQLite

- Built a 19-agent multi-specialist swarm (deterministic compliance, forensics, geospatial) with failure-aware evidence degradation; fused outputs into a composite risk score (35% ML, 25% compliance, 20% anomaly, 10% contractor, 10% evidence); engineered 176 canonical features (financial ratios, progress rates, contractor concentration); isotonic-calibrated Random Forest: PR-AUC 0.9193, F1 88.15%, 0.66% FPR on 750 observations; implemented WGS-84 spatial collision detection (250m radius), temporal RAG with verbatim MPLADS 2023/GFR 2017 statutory citations, and SHA-256 chained tamper-evident audit ledger; maintained 536 passing pytest tests across security RBAC, end-to-end pipeline, failure injection, and agent sensitivity suites.

THEODRA: Workplace Harassment Early-Warning System — GitHub
Python, Isolation Forest, Flask, TypeScript, Next.js, PostgreSQL

- Constructed a metadata-only Isolation Forest anomaly detection over synthetic communication metadata to surface behavioral anomalies for Internal Committee (IC) review; never touching message content; k-anonymity layer (k=5) generalizes quasi-identifiers before any pattern reaches human review, preventing re-identification; anonymous reporting portal with no persistent identifiers and aggregate-only IC dashboard; multi-party sign-off required for individual escalation.

SHURI-LAN: Deployed IoT Threat Detection System — GitHub
Python, Scapy, Nmap, Kali Linux, Machine Learning

- Developed a lightweight IoT scanner running fully on a laptop (no cloud dependency): device fingerprinting, open-port detection, real-time anomaly flagging; in port-scan and unauthorized-connection tests, decoy Telnet/MQTT/HTTP/SSH services captured 100% of probe traffic, logging source IP, timestamp, and targeted service for dashboard review; validated firewall-aware risk scoring (filtered vs open ports): cut an IP camera's blast-radius score 10→7 when Telnet was firewalled; a distinction standard IoT scanners miss.

TECHNICAL SKILLS

- **Languages:** Python, SQL, Bash, JavaScript, TypeScript, C, C++, Java
- **Security:** Scapy, Nmap, Kali Linux, Wireshark/Tshark, MITRE ATT&CK, Threat Modeling, IoT Security, Network Intrusion Detection
- **ML & Data:** Scikit-learn, PyTorch Geometric, SHAP, GNNs (GraphSAGE), SMOTE, Conformal Prediction, Isolation Forest, Anomaly Detection, Adversarial ML
- **Web & Backend:** Flask, FastAPI, Next.js, React, Streamlit, SQLite, PostgreSQL, Redis
- **Tools:** Git, Linux, Jupyter, Docker, Vercel

ACHIEVEMENTS & EXTRACURRICULARS

Hackathons:

- *Smart India Hackathon 2026 — Internal Round Qualifier ×2*
- *UNNATI (AI scholarship & fellowship management for Scheduled Tribes), Sanchay-AI (AI MPLADS anomaly & fraud detection)*
- *Datathon 2026 — TrustSentinel (UPI fraud detection, per-card behavioral profiling)*
- *Hackhouse Goa — ProofOfFace (privacy-preserving facial discovery + blockchain attestation)*
- *Terrier Cyber Quest 2026, AI Kavach — VARUNA (autonomous cyber reasoning)*
- *Velora 1.0 — SHURI-LAN (deployed IoT threat detection)*

Certifications:

- *Cisco — Junior Cybersecurity Analyst Career Path, Data Analytics Essentials*
- *NASSCOM MeitY CoE — Digital Engineering 101*
- *IBM SkillsBuild — AI Fundamentals*

EXPERIENCE & LEADERSHIP

Management Team Member — GeeksforGeeks Campus Chapter, MITAOE — **2025 – Present**
Part of the core team organizing technical events, workshops, and coding initiatives for a community of 200+ students